

Programming in Python 2025/26

Assignment 02

may 2026

Topic: Control Structures

INSTRUCTIONS

This Assignment covers **Control Structures Chapter**. Each question should be answered in a different file. You should submit the python file and a **pdf** file where you have captured the outputs (ONLY OUTPUTS) after running the questions solutions.

Submission deadline: Latest by sunday 31th May 2026 at 1600 hrs. Submit through your cr.

Question 01

1. A palindrome is a number or a text phrase that reads the same backwards or forwards. For example, 12321, RACECAR are palindromes. Write a program that reads in a **positive** five-digits **integer** and determines whether it is a palindrome. The program should restrict the user to enter positive five-digits integer. If not, the user should be given an error message: Entered integer ... is not a five digit integer, try again. If number is not positive, the user should be given an error message: Entered integer ... is not positive, try again and then re-prompted to re-enter the integer. Sample display:

```
Enter five-digits integer: .....
```

```
Entered integer is pandrome / not palindrome
```

Test your program with following integers

(a) 345 (b) -65456 (c) 65456 (d) 45678 (e) 12321

Question 02

- (a) Write a program using **for loop** to calculate and display in 6 decimal places the sum of the first N terms of the following series where N is a positive integer (N>0) specified by user. You can also include **while** loop in restricting N value to be greater than zero but **for loop** must appear in your code. When the program is run, user will be prompted to enter value of N.

$$S_n = \frac{1}{2} + \frac{2}{3} + \frac{3}{4} + \frac{4}{5} + \dots$$

Test your program with: (i) N=0 (ii) N=1 (iii) N=15 (iv) N=100 (v) 1000

(b) Repeat (a) using while loop(s)

(c) Using nested loops of your choice, write a program to generate the following pattern:

```
1 2 3 4 5 6
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

Question 03

Write a program that reads an unspecified number of positive and negative integers (one number at a time). Reading of inputs ends when the input is 999 and 999 should not be counted (it is used as `sentinel` to signal the end of the input). Zero (0) should be treated as positive.

The program should determine and display how many positive and negative values have been read, average of positive numbers, average of negative numbers, maximum of positive numbers, and maximum of negative numbers. Display the average to two decimal places. If there is no any positive number or negative number then display the message "No any Positive number has been entered" or "No any negative number has been entered" when you are displaying info.

If the user enters 999 **only**, the program should display the message "You didn't enter any number, program terminated" and then the program terminates.

Sample display:

```
Enter an integer, end inputs by 999: 1
Enter an integer, end inputs by 999: -23
Enter an integer, end inputs by 999: 47
Enter an integer, end inputs by 999: 94
Enter an integer, end inputs by 999: -34
Enter an integer, end inputs by 999: 999
```

```
==== Positive numbers info ====
The number of positives is:      3
The sum of positives is:        142
The average of positives is:    47.33
Maximum value of positives is:  94
```

```
==== Negative numbers info ====
```

The number of negatives is: 2
The sum of negatives is: -57
The average of negatives is: -28.50
Maximum value of negatives is: -23

Test your program with following data:

- (a) 15, 22, 49, -9, 57, -45
- (b) 23, 199, 89, 12, 0
- (c) -23, -199, -89, -12
- (d) -12, 98, -42, -38
- (e) 999

Question 04

Write a **menu** driven program to enable user to issue commands. When the program is run it displays the following menu:

Following services are available:

1. Start the motor
2. Stop the motor
3. Display help
4. Exit

Enter your choice (1..4):

Choice no 1 displays the message: Motor started

Choice no 2 displays the message: Motor stopped

Choice no 3 displays the following three lines:

Press 1 to start the motor

Press 2 to stop the motor

Press 4 to exit the program

If choice is outside the range, then error message Wrong choice, please re-enter your choice should be displayed and use re-enters another choice.

The program continues to run until the user enters 4 when the program terminates and displays the message Program Terminated

Extra: The question can be improved. If user presses 2 and the motor is not yet started, the message should be displayed that the motor is not yet started.